Computers and Education: Artificial Intelligence 10 (2026) 100548

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Predicting student performance: A comprehensive review of machine learning, deep learning, and explainable AI approaches

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HIGHLIGHTS

- A systematic review of ML/DL techniques for predicting student performance.
- PRISMA methodology was used to select 72 studies from an initial pool of 281.
- OULAD and UCI datasets are the most commonly employed in this domain.
- Tree-based models and neural networks dominate predictive approaches.
- Explainable AI (XAI) is emerging as a key component to support pedagogical decisions, although its adoption remains limited.

ARTICLE INFO

Keywords:
Machine learning
Deep learning
Educational data mining
Student performance prediction
Learning analytics
Explainable artificial intelligence (XAI)
Evidence-based teaching

ABSTRACT

The application of Machine Learning (ML) and Deep Learning (DL) in Educational Data Mining (EDM) is revolutionizing the educational field. Researchers have been particularly interested in predicting student performance at an early stage. These early predictions can significantly benefit students' learning experiences, allowing educators and other stakeholders to plan timely interventions. This review examines studies employing these technologies, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure the approach is transparent and replicable. This systematic review begins with preselecting a total of 281 records that go through a rigorous screening process. Then, 72 retained studies are analyzed to uncover the types of databases utilized, examine the most common features and labels in predictive models, and identify the prevalent ML and DL methods and the reasons for their selection. In addition, the review highlights the role of explainability in making complex models more interpretable and pedagogically meaningful. However, only a limited number of studies explicitly connect these predictive approaches to educational innovation. This review therefore emphasizes how predictive and explainable AI (XAI) can bridge that gap by supporting evidence-based teaching, adaptive learning, and more equitable decision-making in education.

1. Introduction

In recent years, advancements in information technology have significantly impacted various sectors, including education (Wang et al., 2024). Educational institutions now gather vast amounts of data that can be analyzed to improve students' academic trajectories and overall learning experiences. One promising area of research that utilizes this data is student performance prediction, which seeks to identify patterns and factors influencing academic success. Accurate predictions can assist educators in tailoring interventions, providing targeted support, and enhancing educational outcomes (Burgos et al., 2018).

The ability to predict student performance enables early interventions, which can positively affect students' academic paths. Research has shown that predicting student grades allows for the implementation of effective tutoring action plans for those in need (Tondeur et al., 2008), leading to a notable reduction in dropout rates. Pre-tertiary and higher education institutions increasingly recognize that early and accurate forecasting of student performance is crucial for academic success.

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https://doi.org/10.1016/j.caeai.2026.100548
Received 20 July 2025; Received in revised form 18 January 2026; Accepted 21 January 2026
Available online 23 January 2026
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Timely predictions are essential for identifying and addressing learning challenges promptly.

Educational Data Mining (EDM) and Artificial Intelligence (AI) have introduced advanced analytical tools to this field (Sarker et al., 2024). By applying these technologies, educators can take reactive and proactive approaches to support students, thereby improving academic outcomes. Predicting student performance involves analyzing different factors, such as demographic, personal, academic, behavioral, psychological, and socioeconomic factors. This comprehensive approach allows institutions to implement timely and effective measures to address students' needs.

Machine Learning (ML) and Deep Learning (DL) have become powerful tools in this domain (Zhang et al., 2025). Predictive models using these tools can uncover complex relationships and trends that traditional analytical methods might miss. They can adapt and improve, providing deeper insights and more accurate predictions. EDM and AI techniques are becoming common in analyzing data from diverse sources, such as face-to-face classes, online learning platforms, Learning Management Systems (LMS), educational software, and standardized tests. However, the opacity of some predictive models can make it challenging to identify the factors influencing the predictions, emphasizing the need for explainable models.

Several prior reviews have contributed to understanding student performance prediction: (Sekeroglu et al., 2021) provided an overview of ML techniques, (Alnasyan et al., 2024b) examined DL approaches, (Alamri and Alharbi, 2021) explored explainable models, and (Linardatos et al., 2020) synthesized interpretability methods across domains. Extending these foundations, this review systematically examines ML, DL, and Explainable Artificial Intelligence (XAI) together, illustrating how these paradigms intersect in enhancing predictive accuracy, interpretability, and educational value.

Guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, this review analyzes 72 peer-reviewed studies (2017–2024) on ML, DL, and XAI in EDM. It examines how datasets, feature engineering, and evaluation metrics shape predictive performance and transparency, with particular attention to how explainability bridges algorithmic complexity with pedagogical insight. By connecting methodological trends with educational implications, the review demonstrates how predictive models and XAI can inform evidence-based teaching, adaptive learning, and equitable decision-making.

2. Methodology

This systematic review was conducted following the guidelines set by PRISMA. It is organized into four phases: planning, where the scope and methods of the review are defined; searching, which involves database identification and the search approach for potential studies; selecting, where studies are chosen on the basis of the predefined inclusion and exclusion criteria; and finally extracting information for discussion as well as synthesizing the relevant references for this study.

2.1. Planning

The planning phase begins with the formulation of precise research questions (RQs) to direct the scope and objectives of the review:

- RQ1: What types of datasets are commonly used for predicting student performance and what are their characteristics?
- RQ2: What features and labels are used to predict student performance?
- RQ3: Which machine learning and deep learning methods are most frequently used to predict student performance in educational research, and what are the reasons for their selection?
- RQ4: How are interpretability and explainability employed in the studied literature?

These RQs guided the choice of the databases and shaped the search strategy in order to find the most relevant literature. The search process clearly defines which studies are included or excluded in the systematic review. Each step of this process is explained to ensure transparency in the methods used.

2.2. Source databases and search approach

Full text articles related to ML and student performance prediction in English from 2017 to 2024 were searched on ScienceDirect, IEEE Xplore, and Scopus. To effectively gather pertinent articles, multiple rounds of searches were conducted and diverse search query formulations were employed. These included the use of key terms, Boolean logic, and wildcard operators depending on the source database particularities. The queries were strategically combined and implemented to guarantee the acquisition of articles relevant to the topic.

The search query employed across the three databases is the following: ("Machine Learning" OR "Deep Learning") AND Student AND ("Performance Prediction" OR "Grade prediction" OR "Success prediction") AND "data mining"

2.3. Selection process

To meet the research goals, a systematic approach was designed for study selection. All stages of screening and data extraction were conducted by a single reviewer, following PRISMA guidelines. A preliminary search was conducted across databases using the developed search query to identify relevant studies. These were imported, organized, and cleaned in Mendeley, with duplicates eliminated to ensure uniqueness. The selection process proceeded with screening of titles and abstracts, followed by full-text review to exclude studies not meeting the predefined inclusion and exclusion criteria in Table 1. Each stage was performed iteratively and cross-checked multiple times, minimizing the risk of omission or selection bias and ensuring reproducibility.

3. Supervised machine learning

The most commonly used ML models in the reviewed work are Supervised Machine Learning (SML) models. SML models play an important role in EDM as they allow researchers to predict predefined labels based on a set of features, also called attributes. Both labels and features are selected on the basis of their relevance to the research focus. The choice of SML algorithms depends on the dataset's nature and complexity. In this section, algorithms used in the reviewed articles are explored and categorized based on their learning mechanisms.

Throughout this review, we organize models based on their prediction task: regression tasks predict continuous outcomes (e.g., GPA, exam scores), while classification tasks predict categorical outcomes (e.g., at-risk vs. not at-risk). When we refer to "regression" in this review, we encompass both classical statistical approaches (e.g., linear regression) and learning-based regression methods within ML and DL frameworks. This task-based organization reflects current practices in EDM, where the prediction objective defines the analytical approach.

Table 1
Inclusion and exclusion criteria for record selection.

Inclusion criteria	Exclusion criteria
Peer-reviewed journal articles	Review, proceedings, and books
Research based on predicting students' academic performance	Research not focusing on student performance prediction
Research employing supervised ML or DL	Research not employing supervised ML or DL
Articles published between 2017 and 2024	Research lacking sufficient methodological details or that did not involve empirical experimentation
English language written articles	Papers with duplicate entry

3.1. Linear models

Linear models establish a linear relationship between a dependent variable and one or more independent variables. They are widely used in ML because of their simplicity, interpretability, and computational efficiency. In this review, linear models bridge traditional regression theory and modern predictive modeling. Classical formulations such as Multiple Linear Regression (MLR) and Logistic Regression (LR) rely on well-structured, smaller datasets and strict statistical assumptions. Their ML extensions, however, relax these constraints through techniques such as regularization and stochastic optimization, enabling scalability to larger and less structured educational datasets. Decision Trees (DTs), although non-linear, can address both regression and classification tasks, serving as a flexible and interpretable alternative. Linear models thus act as interpretable baselines and adaptable components within broader ML and DL frameworks.

3.1.1. Multiple linear regression (MLR)

MLR is a commonly used LM to predict continuous outcomes (Sandoval et al., 2018), such as a student's final grade for instance and is expressed by (1).

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \epsilon \tag{1}$$

Where:

- y is the dependent variable.
- β_0 is the intercept term.
- $\sum_{i=1}^{n} \beta_i x_i$ represents the sum of the products of the coefficients (β_i) and the independent variables (x_i).
- ϵ is the error term.

3.1.2. Logistic regression (LR)

LR defined by (2) is utilized for binary classification tasks, such as predicting whether a student passes or fails. The logistic function calculates a value between 0 and 1 that represents the probability that the dependent variable belongs to a specific category (Riestra-González et al., 2021).

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + \exp\left(-\left(\beta_0 + \sum_{i=1}^{n} \beta_i x_i\right)\right)} \tag{2}$$

where:

- y is the dependent variable.
- β_0 is the intercept term.
- β_i are the coefficients for the predictor variables.
- x_i are the predictor variables.

3.2. Tree based models

Tree based models algorithms use a series of decision rules to make predictions or classifications. They are represented by a tree structure. Its branches represent decision paths, and its leaves represent outcomes. Common tree-based models used in ML and EDM include decision trees, random forests, and gradient boosting machines.

3.2.1. Decision trees (DT)

DTs are the most basic form of tree-based models. They serve as the foundation for more complex algorithms. A DT uses a simple, hierarchical decision-making process. It splits the data into branches at each node based on specific feature values. This process continues until a leaf node is reached, which provides the output decision as shown in Fig. 1. DTs are easy to understand and interpret but can be prone to over-fitting if not properly pruned (Waheed et al., 2023).

3.2.2. Random forests (RF)

RF, as shown in Fig. 2, is an ensemble method also referred to as bagging, which builds multiple DTs during training. To enhance diversity and reduce over-fitting, RF employs bootstrapped data sampling and random selection of features for each tree. For predictions, it aggregates results through majority voting for classification tasks and averages the results for regression tasks. All variations of RF build upon these fundamental principles, and each variation modifies certain aspects of randomness or learning methodology to address specific needs (Hoffait and Schyns, 2017).

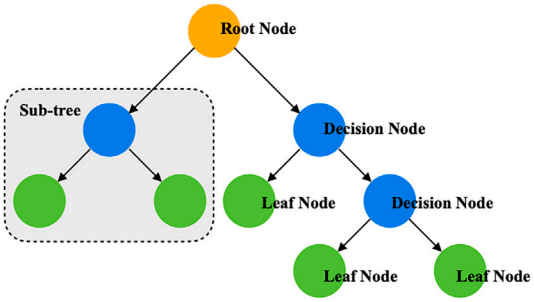


Fig. 1. Decision tree algorithm workflow.

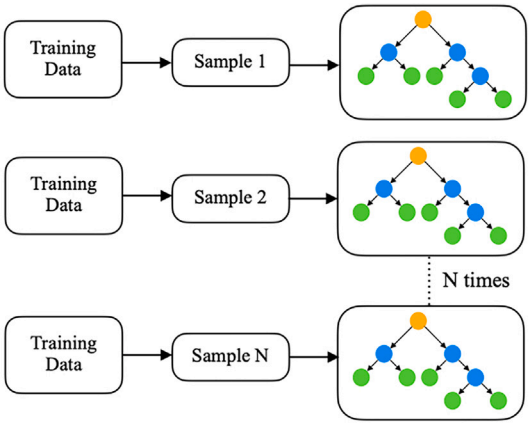


Fig. 2. Random forest algorithm workflow.

3.2.3. Gradient boosting machines (GBMs)

GBMs are an ensemble technique that uses boosting instead of bagging, distinguishing them from RFs. GBMs build the model in a sequential manner, where each new DT corrects the residual errors of its predecessors. This method involves improving the model's accuracy by adding a specified number of DTs, denoted as n as shown in Fig. 3. GBMs require tuning of parameters, including the number of DTs, tree depth, and learning rate, which regulates how strongly each tree attempts to correct the errors of the previous ones (Sahlaoui et al., 2021).

3.3. Support vector machines (SVMs)

SVMs are ML models designed for classification, regression, and outlier detection tasks. They are suitable for both linear and non-linear data. For non-linear data, SVMs utilize kernel functions to transform the data into a higher-dimensional space where a linear hyperplane can effectively separate the classes. The objective of the SVM algorithm is to identify a hyperplane in an N-dimensional space, where N represents the number of features. This hyperplane maximizes the margin, which is the distance between the nearest data points of each class to the hyperplane. This maximization enhances the model's robustness and ability to generalize. Support vectors are the data points nearest to the hyperplane, and they define the margin and influence the hyperplane's position and orientation. Therefore, they shape the decision boundary. The model's sensitivity to classification errors can be adjusted

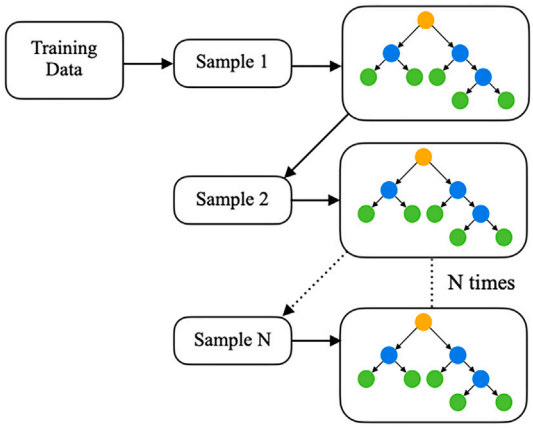


Fig. 3. Gradient boosting machine algorithm workflow.

by the penalty parameter C, which balances the margin width against the misclassification rate on the training data.

Support Vector Regression (SVR) is a variant of SVM designed specifically to predict continuous numerical values, making it ideal for applications such as predicting students' grades. While SVR also aims to find a hyperplane in an N-dimensional space that best fits the data points, it differs from SVM in its approach to error tolerance. SVR includes a margin of tolerance, known as the ϵ-insensitive tube, around the hyperplane. Data points within this tube are considered acceptable errors and do not affect the model. This model aims to fit as many data points as possible within this margin, keeping the hyperplane simple and reducing penalties for data points outside this margin. It makes SVR robust to outliers and effective in predicting continuous outcomes, especially when the relationships between features are complex (Dabhade et al., 2021).

3.4. Bayesian models

The Naive Bayes classifier stands out as the most commonly used Bayesian model in the literature under study. It functions as a probabilistic model that uses the principles of Bayes' Theorem (3). A central assumption of this model is that the features are mutually independent, implying they have no correlation with each other. This simplification makes calculating probabilities easier, so the algorithm works well even with large sets of data.

$$P(y|x) = \frac{P(x|y) \cdot P(y)}{P(x)} \tag{3}$$

where:

- y is the dependent variable
- x is the predictor

3.5. Instance-based models

Instance-based models make predictions by comparing new instances to stored training examples rather than learning generalized rules. The most common instance-based model is k-Nearest Neighbors (k-NN), which classifies or predicts outcomes based on the k most similar training instances using distance metrics such as Euclidean distance. For classification, k-NN assigns the most frequent class label among the k nearest neighbors through majority voting; for regression, it averages their target values.

The choice of k significantly influences performance: small values increase sensitivity to noise, while large values may oversimplify patterns. k-NN is sensitive to feature scaling and irrelevant features, requiring careful preprocessing. While simple and interpretable, k-NN faces computational challenges with large datasets due to distance calculations at prediction time.

3.6. Neural networks (NNs)

NNs have emerged as powerful tools for analyzing large quantities of data, capable of handling diverse data types, whether structured or unstructured, continuous or categorical. This adaptability and versatility are inspired by the human brain, where specific conditions trigger neuron activation, leading to various responses. A neural network consists of layers of interconnected artificial neurons that are activated or deactivated by functions. Each neuron in one layer connects to multiple neurons in the next layer through pathways embedded with weights and biases, which are adjustable parameters. Typically, the architecture of an NN includes an input layer that receives data, several hidden layers that process the data using weights, biases, and activation functions, and an output layer that generates the final prediction or classification.

NNs are diverse, each uniquely tailored to address specific types of data and computational tasks. These networks are structured into different categories based on their architecture and functionality, allowing them to excel in various applications. For the purpose of this systematic review, we focus on the categories most commonly referenced in the literature:

3.6.1. Feedforward neural networks (FNNs)

FFNs consist of layers where connections between neurons do not form cycles, ensuring a straightforward propagation of input data through to the output layer.

The Multilayer Perceptron (MLP) is the most traditional form of FNN, featuring multiple layers of neurons in which each neuron is fully connected to all neurons in the next layer as shown in Fig. 4. It is utilized for a wide range of applications from simple regression to complex classification tasks.

Another variant within this category is the Radial Basis Function Network (RBFN). The latter has one hidden layer, as presented in Fig. 5, whose activation functions are radial basis functions. These functions calculate the distance from the input to a designated center or data point. This is useful for predictive and classification tasks, which depend on how close inputs are to predefined centers.

Finally, Fuzzy Neural Networks (FNNs) are neural networks that incorporate elements of fuzzy logic into the traditional neural network framework. Unlike standard neural networks that operate with fixed binary or continuous outputs, FNNs handle ambiguity and partial truths by integrating fuzzy logic principles, which can lead to more nuanced decision-making and reasoning capabilities.

3.6.2. Convolutional neural networks (CNNs)

CNNs are a specialized type of deep neural network used primarily for processing data that has a grid-like topology like images. They are highly efficient in tasks that involve image recognition, classification, and analysis. A CNN typically comprises convolutional layers, pooling

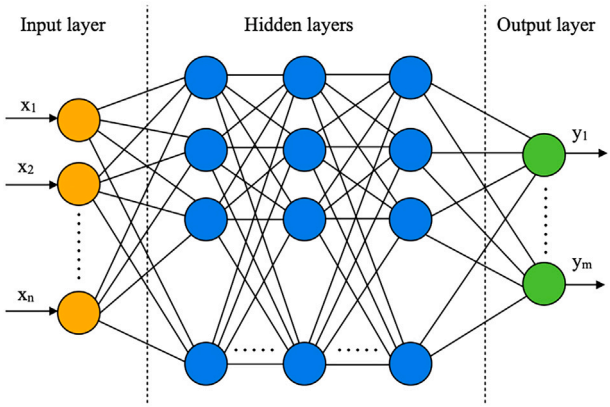


Fig. 4. Multiple layer perception architecture.

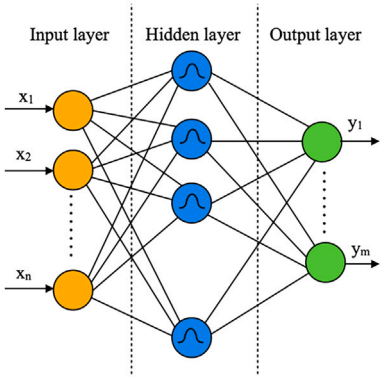


Fig. 5. Radial basis function network architecture.

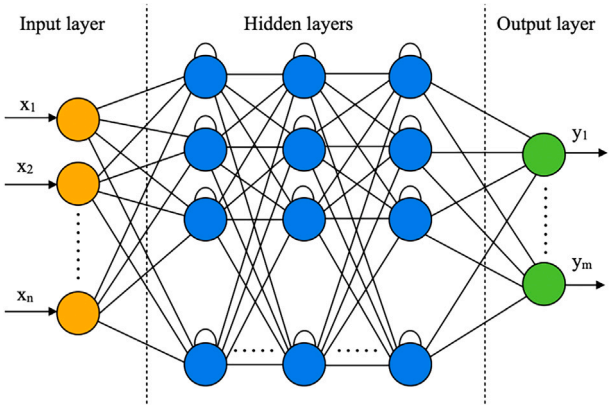


Fig. 6. Recurrent neural networks architecture.

layers, and fully connected layers. In the convolutional layers, filters (also known as kernels) are applied to the input data through a mathematical operation called "convolution". This process produces a feature map that highlights important characteristics of the input, such as edges, textures, or patterns. The pooling layers merge the features collected from their preceding convolutional layer, which reduces the dimensions of feature maps. Finally, the fully connected layers allow for high level feature learning and combination since every neuron in one layer is connected to all the neurons of the next layer.

3.6.3. Recurrent neural networks (RNNs)

RNNs are a type of deep neural network that is specially designed for sequential data. Unlike feed-forward neural networks in which information flows in one direction, RNNs have looping connections that allow the retention of information from previous inputs as shown in Fig. 6. This enables the network to make decisions based on both the current input and the previous ones. Another characteristic of RNNs is that they share the same weight parameters across all layers; yet these parameters are still adjusted through the process of gradient descent and backpropagation.

Long Short-Term Memory networks (LSTM) are a variation of RNNs that were developed to address the long-term dependency issue. In other words, the traditional RNNs often struggle to make accurate predictions when the current decision is not affected by the recent past. LSTMs are designed to remember information for longer periods and therefore remediate the discussed issue.

3.6.4. Hybrid and specialized networks

Hybrid and specialized networks are advanced architectures of neural networks intended to tackle specific types of data. These networks combine different architectures and may also incorporate knowledge that is specific to a particular domain. This is to enhance the overall performance of the network by addressing the challenges of complex data structures.

Graph Neural Networks (GNNs) are an example of such specialized networks. They are used in domains where data-points are interconnected through complex relationships. GNNs operate by applying convolutional techniques to graphs, which allows them to learn from the graph topology itself. This means they can effectively capture both node and edge features and their respective connections.

4. Models' evaluation metrics

To evaluate the effectiveness of the models, the studies use a variety of metrics. In this section, we define the most commonly used ones for both classification and regression models. These metrics provide a standardized way to compare model performances, facilitating a comprehensive analysis during the results and discussion phase. By employing these metrics, we ensure a consistent and objective evaluation of the models' capabilities and limitations.

4.1. Classification models evaluation metrics

For classification tasks, the most commonly used metrics are accuracy, precision, recall, and F1-score. These metrics are derived from the confusion matrix, which details how the model's predictions match the actual outcomes.

4.1.1. Confusion Matrix

The confusion matrix provides a comprehensive overview of the model's performance by showing both correct and incorrect predictions. It applies to both binary and multi-class classification. It includes four components:

- True Positive (TP): Correctly predicted positive instances
- True Negative (TN): Correctly predicted negative instances
- False Positive (FP): Incorrectly predicted positive instances
- False Negative (FN): Incorrectly predicted negative instances

For binary classification, it is relatively straightforward. However, as the classification task scales to multiple classes, the complexity of the confusion matrix increases. When the matrix expands, it becomes more challenging to interpret at a glance. To address this complexity, some studies use heatmaps, which employ color gradients to visualize the frequency of predictions in each cell of the matrix, providing an intuitive understanding of the model's performance. Importantly, the confusion matrix serves as the foundation for calculating key performance metrics, namely accuracy, precision, recall, and F1-score.

4.1.2. Accuracy

Accuracy is the ratio of correctly predicted instances to the total instances as defined by (4).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

4.1.3. Precision

Precision is defined as the ratio of true positive predictions to the total number of positive predictions made by the model. It is calculated using (5).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

4.1.4. Recall

Recall, also called sensitivity, is the ratio of true positive predictions to the total number of actual positive instances and is defined by (6).

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

4.1.5. F1_score

F1_score is the harmonic mean of precision and recall as expressed in (7). It reaches its best value at 1 and its worst at 0. It is particularly useful when it is needed to balance the precision and recall of a test.

$$\text{F1_score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

4.2. Regression models evaluation metrics

While for regression tasks, the most commonly used metrics for evaluating the model are R^2, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Average prediction Accuracy (APA), and Percentage Accurate Predictions (PAP).

4.2.1. Coefficient of determination R^2

The R^2 metric, or coefficient of determination, quantifies how much of the variation in the dependent variable can be explained by the independent variables. It serves as a measure of model fit and is computed as shown in (8).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{8}$$

where:

- y_i are the actual values,
- $\hat{y}_i$ are the predicted values,
- $\bar{y}$ is the mean of the actual values,
- n is the number of observations.

4.2.2. Mean absolute error (MAE)

As shown in (9), the Mean Absolute Error (MAE) represents the average size of the prediction errors, regardless of their direction. It is computed by averaging the absolute differences between predicted and actual values across the test set, giving equal importance to each error.

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{9}$$

where:

- y_i are the actual values,
- $\hat{y}_i$ are the predicted values,
- n is the number of observations.

4.2.3. Root mean square error (RMSE)

RMSE is the square root of the average of squared differences between predictions and actual observations. It penalizes larger errors more than MAE and is useful when large errors are particularly undesirable.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{10}$$

where:

- y_i are the actual values,
- $\hat{y}_i$ are the predicted values,
- n is the number of observations.

5. Results and discussion

5.1. Search results and article selection

The initial search, conducted with the developed search query, yielded a total of 281 records. Upon the removal of duplicates, 270 records remained. A further screening based on titles, abstracts and access to full texts led to the exclusion of 173 records, leaving 97 for full text evaluation. Following a comprehensive full-text evaluation, 22 records that did not meet the inclusion criteria were eliminated. Subsequently, 72 articles were included in the systematic review of the literature. The screening and selection process is summarized in Fig. 7.

Fig. 8 shows the evolution of researchers' interest in the subject. It can be deduced that the research area is relatively new and rapidly growing. The data for 2024 is not included as the search was conducted in January 2024. In fact, EDM, when coupled with ML and DL, is increasingly recognized for its potential to revolutionize the field of student performance prediction in order to improve student outcomes. As more researchers explore the capabilities of these technologies combined, the volume of publications and studies is expected to rise.

The educational setting of the reviewed articles included both pre-tertiary students (n = 15) - comprising secondary and high school students- and higher education (n = 54) meaning university students. Articles that failed to explicitly mention the educational setting have been omitted from Table 2. The studies focusing on university students represent 78%, significantly outnumbering those on pre-tertiary education, which accounted for about 22%. This disparity is likely due to greater access to university databases as open sources, facilitating research at the tertiary level.

Tables 3 and 4 report the frequency of ML and DL models used in the reviewed studies. To improve readability, the corresponding references for each model category are provided in the Appendix (Table 7). Because some studies evaluate multiple algorithms, a single study may appear in more than one model category. Overall, traditional ML models are reported approximately three times more frequently than DL models in the reviewed literature, which likely reflects their earlier and broader adoption in educational data mining, while DL approaches have gained prominence more recently.

5.2. Datasets sources and characteristics

The quality of the dataset is crucial for producing accurate and reliable models. A high-quality dataset ensures that the predictive models can effectively capture the complex factors influencing academic success. Comprehensive data that includes demographic, academic, behavioral, and socio-economic indicators allow for precise predictions. Therefore, careful selection and preprocessing of datasets are essential for developing robust models that provide actionable insights for educational interventions

It is worth noting that many studies are evaluated using a single dataset corresponding to a specific institutional context. This is a common characteristic in EDM research, as data availability is often constrained by privacy and access conditions. Nonetheless, this reliance on single-institution datasets naturally influences the extent to which results can be generalized across different educational settings, and this aspect is taken into consideration in the subsequent analysis.

5.2.1. Datasets sources

Educational datasets used for student performance prediction come from various sources, including publicly available datasets and institutional or proprietary datasets. These datasets offer different levels of access and detail, and they play a critical role in fostering research, enabling reproducibility, and supporting comparisons across studies.

Open University Learning Analytics Dataset (OULAD)

One of the most extensively used open-source datasets is the Open University Learning Analytics Dataset (OULAD), from the Open University, an e-learning institution. The dataset has been employed in various studies, enabling researchers to investigate key factors influencing student success and engagement. OULAD covers four main aspects of student data: registrations, assessments, VLE interactions, and demographics, offering a comprehensive view of student learning behavior.

In the studied literature, eight studies have explored different facets of the OULAD dataset (Al-Zawqari et al., 2022; Esteban et al., 2021;

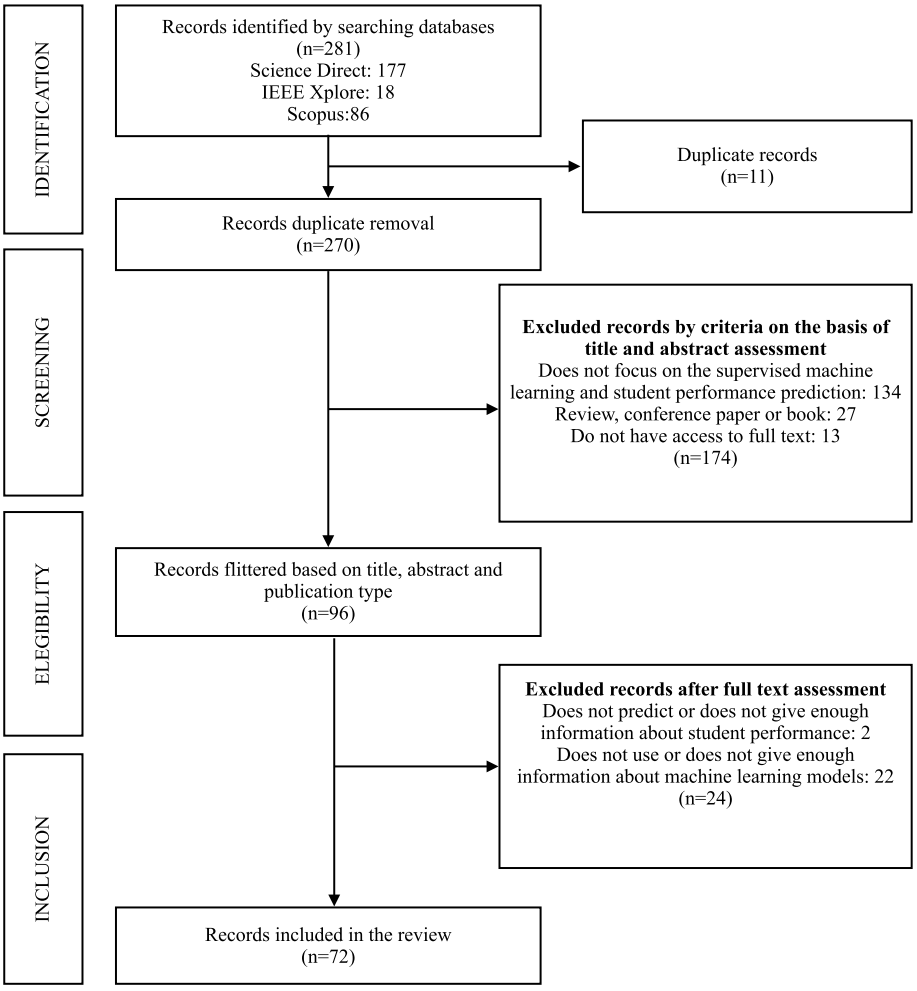


Fig. 7. PRISMA flow diagram for papers' selection process.

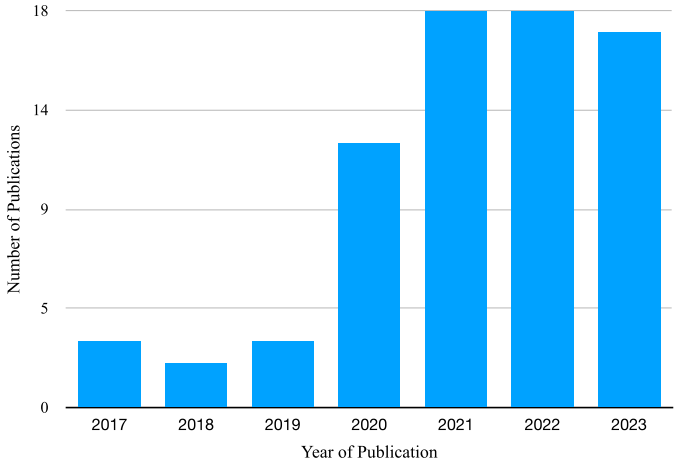


Fig. 8. Number of publications by year.

Table 2
Educational setting of reviewed studies.

Educational setting	Number of studies
Pre-tertiary education	15
Higher education	54

Table 3
Machine learning models used in the reviewed literature and their frequency.

Model type	Model sub-type	Freq.
Linear models	Multiple Linear Regression	4
	Logistic Regression	26
Tree-based models	Decision Trees	30
	Random Forests	31
	Gradient Boosting Machines	26
Support vector models	Support Vector Machine	25
	Support Vector Regression	1
Bayesian Models	Naive Bayes Classifier	21
Instance-Based Models	k-Nearest Neighbour	26

Liu et al., 2022, 2023; Nanavaty and Khuteta, 2024; Rizvi et al., 2019; Waheed et al., 2020, 2023). For instance, (Liu et al., 2023) concentrated on analyzing student demographics and interactions with the Virtual Learning Environment (VLE), while (Al-Zawqari et al., 2022) utilized the full dataset, incorporating student registrations, assessments, VLE interactions, and demographics. The open access and versatility of OULAD make it a valuable resource in EDM, enabling researchers to study diverse aspects of student behavior and performance, and apply it across a range of predictive models.

Table 4
Deep learning models used in the reviewed literature and their frequency.

Model type	Model sub-type	Freq.
Neural networks	Multilayer Perceptron (MLP)	20
	Convolutional Neural Network (CNN)	9
	Radial Basis Function Network (RBFN)	4
	Fuzzy Neural Networks	2
	Recurrent Neural Network (RNN)	3
	Long Short-Term Memory (LSTM)	5
	Hybrid and Specialized Networks	4

University of California, Irvine Machine Learning Repository (UCIMLR)

Another widely used public dataset is the University of California, Irvine Machine Learning Repository (UCIMLR) dataset for Mathematics and Portuguese students. This dataset contains student grades, demographic, social, and school-related attributes, which are collected from school reports and questionnaires. It has been used in five studies (Alnasyan et al., 2024a; Aslam et al., 2021; Begum and Padmannavar, 2022; Fan et al., 2023; Yousafzai et al., 2021), allowing researchers to explore the relationships between student performance and various factors such as demographic characteristics and academic history.

Proprietary Datasets

In addition to publicly available datasets, many researchers use proprietary datasets collected internally by educational institutions, government agencies, and specialized organizations. These datasets often include detailed information that is specific to the institution or program, such as grade reports, learning management system (LMS) logs, and student information systems (SIS). Proprietary datasets are usually more comprehensive than public datasets, offering insights into student behaviors, performance, and interaction with course materials, but they are not readily accessible to the broader research community.

For instance, the United States Air Force's Specialized Undergraduate Pilot Training (SUPT) dataset (Jenkins et al., 2022) is an example of a proprietary dataset that has been used to explore performance-related factors in specialized training environments. These datasets are often used to develop predictive models tailored to specific educational contexts and are invaluable for in-depth analysis of unique student populations. However, because they are not openly available, they do not provide the same level of reproducibility or comparability as public datasets.

5.2.2. Datasets sizes and structure

Educational datasets are complex, capturing various aspects of the student learning experience. They are often longitudinal, tracking students' data over extended periods, which helps identify changes and patterns in their academic journeys. These datasets may include components such as student demographics, academic performance, behavioral data, parental involvement, special education needs, and learning environments.

Educational datasets vary in size and format. Typically, rows in these datasets represent students, with some datasets having multiple rows per student. For example, study (Alangari and Alturki, 2020) includes data on 5566 students over seven years, with each student having 50 or more rows, one for each course. A student may have multiple rows for a single course if they withdrew failed or retook it. Thus, this dataset contains 301,078 examples. Similarly, study (Dien et al., 2020) involves a dataset of 3,828,879 records for 83,993 students and 4699 courses. Another study's dataset includes 589,540 performance records from 2678 courses taken by 14,264 students over 16 semesters (Khan et al., 2021a). Although it lacks a detailed description of the data structure, it can be deduced that the dataset is a student-course combination.

Fig. 9 presents a scatter plot illustrating the dataset sizes used in the reviewed studies. To enhance visualization, the plot includes only datasets with fewer than 4000 instances, as more than 74% of the datasets fall within this range. Some studies utilize multiple datasets, and each point on the plot represents a single dataset. Studies that do not explicitly report dataset sizes are excluded from the visualization.

The five smallest datasets in the reviewed studies contain between 113 and 129 instances (Injadat et al., 2020a,b; Sultana et al., 2017; Tsiakmaki et al., 2020; Xue and Niu, 2023). Conversely, the five largest datasets range from 90,000 to over 3.8 million instances (Alangari and Alturki, 2020; Dien et al., 2020; Hussain and Khan, 2023; Khan et al., 2021a; Sandoval et al., 2018). The variation in dataset sizes has significant implications for model performance. Smaller datasets, are easier to process but may lead to overfitting and limited insights due to insufficient representation of diverse learning patterns. This issue is particularly critical for DL models, which require large amounts of data to leverage their full potential. On the other hand, larger datasets provide more comprehensive patterns but require increased computational resources, making their processing and optimization more complex.

5.2.3. Datasets content

An examination of the features used across the reviewed studies shows substantial variation in the types of attributes employed for predicting student performance. To structure this analysis, we categorized features into temporal and non-temporal groups. Temporal features are time-dependent and typically extracted from logs or sequential academic records, while non-temporal features are static and available prior to enrollment.

Fig. 10 provides a compact overview of the frequency of feature categories used across all reviewed studies, offering a visual synthesis of the feature landscape. The complete detailed classification is presented in Appendix Table 8.

Most studies combined both temporal and non-temporal variables, with temporal academic and behavioral attributes emerging as the most prevalent. This trend reflects the increasing reliance on sequential engagement data and continuous academic records to capture dynamic patterns of learning behavior.

a. Temporal features

Temporal features include variables related to academic progress (e.g., grades over time, assessment results) and behavioral interactions (e.g., learning platform usage, resource access, or participation metrics). Several studies have used both academic and behavioral temporal features (Alachiotis et al., 2022), (Al-Zawqari et al., 2022), (Fan et al., 2023), (Imran et al., 2019), (Sultana et al., 2017).

Behavioral temporal features appear in studies that utilize data from learning management systems (LMS) or other platforms that log sequential student interactions. For instance, (Liu et al., 2022),(Mastour et al., 2023), (Nabizadeh et al., 2022) and(Nachouki et al., 2023) employed variables such as resource views, quiz attempts, and assignment submissions over time. Additionally, some studies (Wang et al., 2023) extracted temporal behavioral features from campus systems such as card usage data for example, breakfast frequency and time, or campus card spending patterns, which provide indirect yet informative insight into student routines and engagement.

b. Non-temporal features

Non-temporal features were used in studies that do not include time-dependent behavioral data (Alangari and Alturki, 2020), (Aslam et al., 2021), (Chen et al., 2023), (Dien et al., 2020), (Hoffait and Schyns, 2017) and in studies focusing on early-stage prediction where only static information was available prior to prediction (Zeineddine et al., 2021). These features are grouped into the following three categories:

Demographic: Attributes such as gender, age, and nationality. These were used in studies such as (Abuzinadah et al., 2023), (Dabhade et al., 2021), (Jenkins et al., 2022), and (Yurtkan et al., 2023).

Academic Background: This group includes the school system, grades, placement test scores, enrolled program, and course load. Examples appear in (Alangari and Alturki, 2020), (Begum and Padmannavar, 2022), (Imran et al., 2019), and (Turabieh et al., 2021).

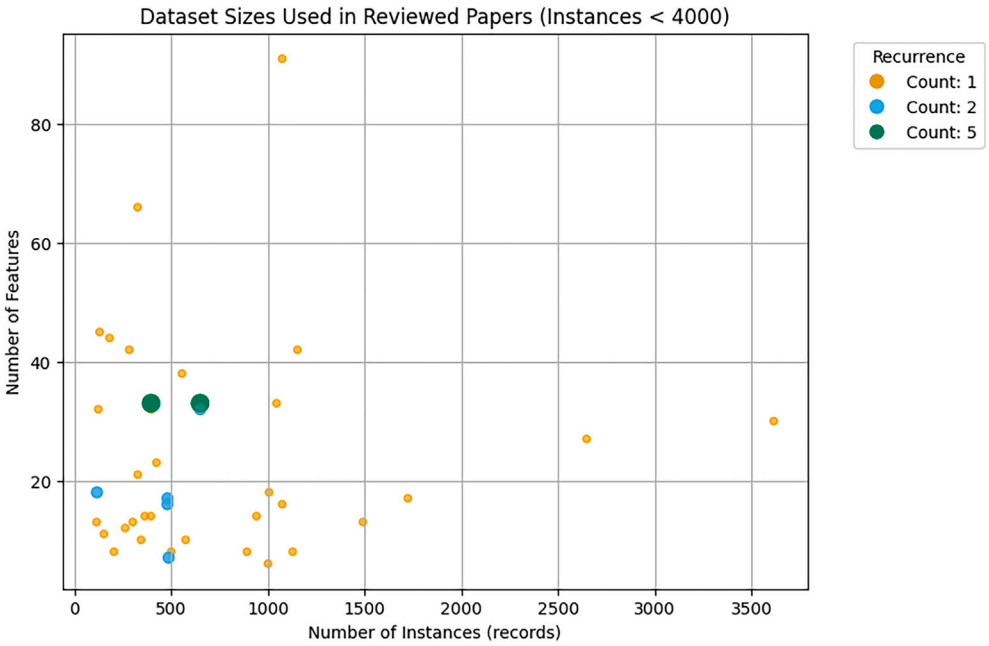


Fig. 9. Datasets sizes and recurrences (Instances < 4000).

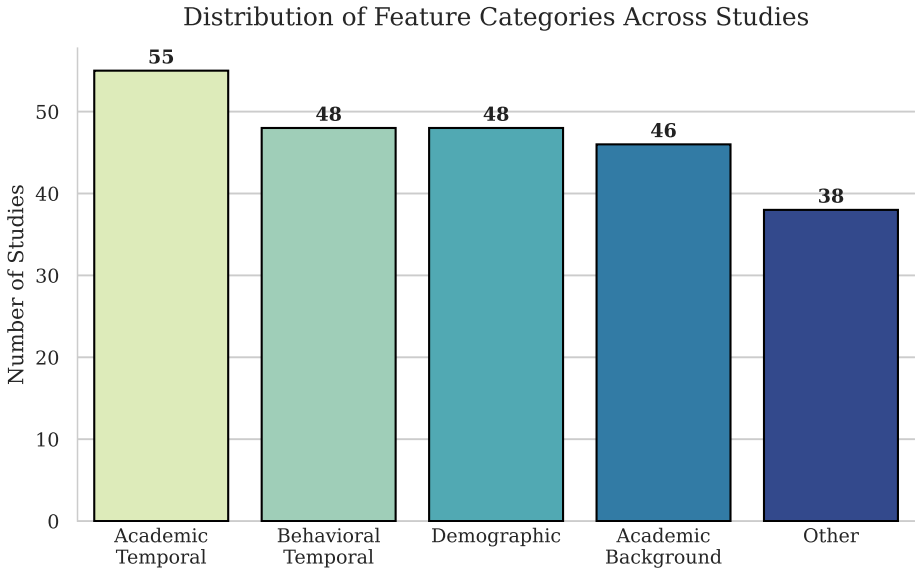


Fig. 10. Distribution of feature categories across the reviewed studies. The figure highlights the predominance of temporal academic and behavioral features, illustrating the growing emphasis on time-dependent learning analytics in the literature.

Other: This category includes a diverse set of features beyond demographics and academic background. It encompasses socioeconomic indicators such as scholarship status (Hoffait and Schyns, 2017), family situation and employment status (Waheed et al., 2023), and parental involvement variables like parental responsibility, school satisfaction, and survey responses (Jasim et al., 2021). It also covers behavioral and contextual factors such as dormitory residence and sponsorship (Khan et al., 2021b), romantic relationships, internet usage, nursery attendance, travel time (Fan et al., 2023; Turabieh et al., 2021), personality traits and policy waivers (Jenkins et al., 2022), future education plans and social activities (Pek et al., 2023), self-concept and leadership (Sultana et al., 2017), administrative response times (Giannakas et al., 2021), and teacher-related contextual features (Wang et al., 2023). These heterogeneous features reflect various external influences on student performance captured in multiple studies.

5.3. Data preprocessing

Data preprocessing is vital in both ML and DL to prepare raw data for model training. Key steps include collecting data, cleaning it by removing noise and inconsistencies, handling missing data, and splitting it into training and testing sets. Feature scaling ensures all features contribute equally to the model. While both ML and DL require these steps, they differ in their specifics. ML often involves manual feature engineering, using domain knowledge to create and select relevant features, and encoding categorical variables into numerical formats. DL models, on the

other hand, learn features directly from the data, reducing the need for manual intervention and often use data augmentation to enhance the dataset.

5.3.1. Data cleaning and transformation

Al-Zawqari et al. (2022), who used the OULAD dataset, undertook data cleaning to address inconsistencies and issues that hindered cross-validation. The authors discovered discrepancies in both the information records and the weights assigned to course assessments. For instance, some students who had withdrawn from a course were still assigned a failing grade, which was subsequently corrected. They also found that the weights of assessments did not match those stated in the dataset documentation, leading them to exclude the assessment file from their analysis. Additionally, missing values for the feature imd_band (Index of Multiple Deprivation band) were identified and imputed to ensure data completeness.

Injadat et al. (2020a) replaced all empty marks in the datasets with a value of 0 to handle missing data and improve classification accuracy. Next, the marks were scaled out of 100 to maintain consistency. Additionally, any decimal values were rounded to the nearest whole number to standardize the data. They also used PCA but only for visualization purposes, not as part of the transformation of data fed into the algorithms.

5.3.2. Data balancing

Class imbalance is one of the most recurrent challenges in EDM. In many learning analytics datasets, the proportion of students classified as "at risk" is far smaller than that of successful learners, which can bias predictive models toward the majority class. Addressing this imbalance is crucial for improving both the sensitivity and fairness of predictive models.

Imbalanced data influences model behavior depending on the analytical framework and prediction objective. In regression tasks, imbalance does not invalidate estimation, though strong distributional disparities can reduce stability. In classification tasks, which dominate student performance prediction, imbalance is more consequential because minority outcomes (e.g., dropout, failure) are precisely what educators seek to identify. Reviewed studies employed strategies ranging from data-level resampling to algorithm-level loss adjustments to preserve meaningful minority-class signals for actionable early warning.

Several studies exemplify this evolution. Abuzinadah et al. (2023) demonstrated that synthetic oversampling through the SMOTE algorithm improves classification performance compared to the original unbalanced dataset. Wang et al. (2023) proposed the Probabilistic Student Academic Performance (ProbSAP) framework, which achieves balance not by adding or removing data points but by refining the representation of underrepresented groups through scalable metadata clustering and robust loss weighting. This probabilistic treatment enhances minority group visibility without distorting the data distribution. Similarly, Hung et al. (2020) avoided statistical resampling altogether and instead redefined the "at-risk" label threshold, creating a more balanced class ratio through a meaningful adjustment of the outcome definition.

Collectively, these approaches demonstrate a shift toward balancing strategies that consider both model characteristics and the pedagogical importance of identifying at-risk students. This transforms imbalance management into a deliberate design decision rather than a generic preprocessing step.

5.3.3. Feature selection and feature engineering

Feature selection is a critical process in both ML and DL, aimed at enhancing model performance by identifying the most relevant features. In traditional ML, feature selection serves to reduce overfitting, improve model interpretability, and boost generalization to unseen data. This process ensures that only the most important variables are retained for the model.

a. Feature Selection and Feature Engineering for ML Models

Imran et al. (2019) applied a filter-based method to rank and select features using information gain. From 33 attributes in the UCI Machine Learning Repository dataset, the top 12 were retained. Among these, second-term grades, first-term grades, number of past failures, absence rate, and father's education level were identified as the most influential features.

Mastour et al. (2023) adopted SHAP (SHapley Additive exPlanations) for feature selection. SHAP was used after training an initial model to compute the contribution of each feature to the prediction task. Features were ranked based on their mean absolute SHAP values, and only the most impactful ones were retained for model development. Biochemistry grades, Anatomical Sciences grades, and GPA had the strongest positive SHAP values, whereas features like General Knowledge grades were less impactful and were potentially excluded.

Hussain and Khan (2023) used a Genetic Algorithm (GA) to optimize the selection of relevant features. Starting from 126 features, the GA iteratively evaluated feature combinations, mimicking natural selection to retain those that improved model performance. This process reduced the feature set to 30 attributes and significantly enhanced the performance of both DT and k-NN models in classification and regression tasks.

Priyambada et al. (2023) employed extensive feature engineering during preprocessing. Raw academic data were transformed into structured features based on learning behavior (e.g., alignment with curriculum schedule: match, before, after) and domain knowledge (e.g., credit and grade distributions across general knowledge, IT, and business courses). Both static (semester-level GPA and credits) and evolving (cumulative GPA and credits) indicators were used to construct a growing feature set that captures student progress over time. To manage the resulting high-dimensional feature space, the authors applied clustering and generalization techniques to reduce dimensionality while preserving predictive value.

Khan et al. (2021a) conducted explicit feature engineering by transforming historical academic records into five structured, high-level attributes to enable early prediction of student performance. The features included student quality (derived from the preceding semester CGPA), domain knowledge (based on performance in semantically similar past courses), expected teaching quality (inferred from historical student evaluations), expected ease of scoring (based on historical grade distributions), and backlog status (measured as the proportion of previously failed courses not yet cleared). These handcrafted categorical features were used to train both traditional classifiers and a novel algorithm proposed by the authors called Random Wheel, which relies on the comparative influence of feature combinations.

b. Feature Selection for DL Models

While DL models, particularly neural networks, inherently perform automatic feature extraction, explicit feature selection remains relevant. Techniques such as dropout and regularization help manage feature importance indirectly by preventing over-reliance on specific inputs. However, studies show that reducing input dimensionality through external methods can still yield performance benefits.

Al-Zawqari et al. (2022) demonstrated that a neural network trained on a reduced set of features outperformed one trained on all available inputs. The authors used a feature selection loop involving DTs and rank correlation analysis to eliminate redundant features, which helped reduce noise and enhance model accuracy.

Olabanjo et al. (2022) applied Principal Component Analysis (PCA) as a feature selection method, reducing the feature space from 31 to 17 variables, encompassing psychomotor, cognitive, and academic performance indicators, before training a Radial Basis Function Neural Network (RBFNN). PCA transformed the original correlated features into uncorrelated principal components based on the eigenvalues of the covariance matrix. Retaining only the top components improved computational efficiency, reduced multicollinearity, and mitigated overfitting.

Additionally, Alnasyan et al. (2024a) proposed a hybrid approach combining Binary Particle Swarm Optimization (BPSO) with a

Convolutional Neural Network (CNN). Their BPSO-CNN framework selected a reduced set of features, which led to a 1.6% improvement in accuracy compared to a traditional CNN, highlighting the effectiveness of tailored feature selection in DL contexts.

5.4. Models' performance analysis

The prediction task, whether classification or regression, fundamentally shapes model selection, evaluation, and interpretability in student performance prediction. Classification tasks predict categorical outcomes (e.g., at-risk vs. not at-risk, pass vs. fail, dropout vs. retention), while regression tasks predict continuous outcomes (e.g., GPA, exam scores). Both ML and DL frameworks accommodate these complementary task types, though classification tasks dominate the reviewed literature, reflecting the field's emphasis on identifying at-risk students for timely intervention.

The models employed are broadly grouped into ML and DL categories, each with various subtypes. Tables 3 and 4 list the most frequently used models across reviewed studies. However, performance comparison differ substantially between task types due to distinct evaluation frameworks: classification tasks use metrics such as accuracy, precision, recall, and F1-score, while regression tasks employ RMSE, MAE, and R^2. Additionally, regression results cannot be directly compared across studies due to differing prediction scales and target variables.

Given these fundamental differences, we analyze classification and regression task performances separately in the following subsections.

5.4.1. Classification: predicting categorical outcomes

Classification tasks in the reviewed literature predominantly address binary outcomes (at-risk vs. not at-risk, pass vs. fail, dropout vs. retention) and multi-class outcomes (letter grades, qualitative performance levels). Performance is evaluated using standard metrics including accuracy, precision, recall, and F1-score, enabling direct comparison across studies. Tables 5 and 6 summarize the highest performance for classification tasks across reviewed studies, organized by model architecture.

Tables 5 and 6 present the highest-performing models from each reviewed study where tabular representation is feasible, revealing distinct performance characteristics across model architectures. Among ML approaches, tree-based models demonstrate the strongest and most consistent performance, with Gradient Boosting Machines achieving exceptional accuracies up to 99.4% (Sahlaoui et al., 2021) and ensemble methods exceeding 97.8% (Alcaraz et al., 2021; Pek et al., 2023). DTs show considerable variation (84% – 99.9%), suggesting high sensitivity to dataset characteristics and preprocessing strategies. Random Forests maintain relatively stable performance (78% – 92%), though with notable variance influenced by feature selection and class balancing techniques. Simpler models exhibit mixed results: Logistic Regression achieves moderate performance, while k-NN demonstrates strong accuracy (93%) when paired with effective dimensionality reduction (Ravi et al., 2023). SVMs show the widest performance range, indicating that kernel selection, hyperparameter tuning, and data preprocessing critically determine their effectiveness in educational contexts.

Deep learning architectures demonstrate competitive but more variable performance patterns. MLPs range from 72.5% to 96.4%, with top-performing implementations leveraging careful hyperparameter optimization and architectural design (Aslam et al., 2021). CNNs achieve up to 96.7% when enhanced with feature selection techniques (Alnasyan et al., 2024a), demonstrating adaptability beyond their traditional image processing domain. Specialized architectures show promising accuracy.

Table 5
Performance summary of classification ML models.

Model type	Model sub-type	Reference	Acc.	Prec.	Rec.	F1-score
Linear models	Logistic Regression	(Jang et al., 2022)	83.8%	79.0%	82.0%	80.0%
		(Alachiotis et al., 2022)	–	88.0%	88.0%	88.0%*
Tree-based models	Decision tree	(Khan et al., 2021b)	85.0%	93.0%	78.0%	91.0%
		(Abuzinadah et al., 2023)	99.9%	90.0%	90.0%	90.0%
		(Hussain and Khan, 2023)	96.7%	–	–	–
		(Imran et al., 2019)	95.8%	95.8%	95.8%	95.8%
		(Alangari and Alturki, 2020)	91.0%	–	–	–
		(Jenkins et al., 2022)	93.7%	94.0%	94.0%	94.0%
		(Sultana et al., 2017)	84.0%	–	–	–
	Random forest	(Alhazmi and Sheneamer, 2023)	87.2%	89.1%*	87.0%*	87.8%*
		(Jasim et al., 2021)	89.0%	90.0%	89.0%	89.5%
		(Hoffait and Schyns, 2017)	92.0%	–	–	–
		(Mastour et al., 2023)	83.0%	97.1%	81.4%	82.3%*
		(Nabizadeh et al., 2022)	78.0%	–	–	–
	Gradient boosting machines	(Sahlaoui et al., 2021)	99.4%	99.5%	99.4%	99.4%*
		(Cheng et al., 2024)	94.2%	94.2%	94.2%	94.1%
		(Begum and Padmannavar, 2022)	98.1%	95.4%	98.9%	96.5%
			96.8%	87.9%	89.1%	88.2%
		(Esteban et al., 2021)	96.6%	94.4%	95.5%	94.6%
			94.4%	91.6%	95.6%	93.5%
	Other ensemble models	(Alcaraz et al., 2021)	97.8%	–	98.8%	–
		(Injadat et al., 2020a)	93.1%	71.0%*	73.0%*	72.0%*
		(Niyogisubizo et al., 2022)	92.2%	93.0%	93.0%	92.0%
		(Pek et al., 2023)	98.4%	96.8%	97.0%	96.5%
		(Priyambada et al., 2023)	83.0%	83.0%	91.0%	87.0%
Bayesian models	Naive Bayes	(Alangari and Alturki, 2020)	91.0%	–	–	–
Instance based models	k-NN	(Ravi et al., 2023)	93.0%	98.0%	95.0%	92.0%
SVM	SVM	(Damuluri et al., 2020)	70.0%	–	–	–
		(Walid et al., 2022)	–	96.4%	98.7%	97.5%
Novel ML models	Random Wheel Algorithm	(Khan et al., 2021a)	80.5%	95.7%	81.2%	86.2%

* Computed from the reviewed article data.

Table 6
Performance summary of classification DL models.

Model type	Reference	Acc.	Prec.	Rec.	F1-score
MLP	(Ranjeeth et al., 2021)	79.3%	79.3%	79.3%	79.3%
	(Riestra-González et al., 2021)	90.0%	87.8%	95.4%	89.4%
	(Sun et al., 2023)	72.5%	82.2%	73.5%	73.6%
	(Aslam et al., 2021)	96.4%	99.2%	93.3%	96.2%
		93.2%	94.9%	90.3%	92.6%
	(Hung et al., 2020)	89.9%	–	–	–
	(Giannakas et al., 2021)	81.8%	74.4%	60.4%	66.7%
CNN	(Alnasyan et al., 2024a)	96.7%	96.8%	96.7%	96.7%
		83.1%	94.3%	93.3%	93.3%
GNN	(Chen et al., 2023)	93.1%	89.3%	93.3%	90.4%
	(Li et al., 2022)	–	92.4%	92.0%	92.2%
LSTM	(Liu et al., 2023)	89.2%	–	–	92.7%
	(Waheed et al., 2023)	84.6%	82.2%	79.4%	80.8%
RBFN	(Olabanjo et al., 2022)	86.6%	86.3%*	93.5%	89.7%*
RNN	(Turabieh et al., 2021)	92.0%	–	–	–
Hybrid models	(Hussain et al., 2020)	96.0%	95.7%	95.3%	95.5%
	(Liu et al., 2022)	92.5%	90.6%	96.5%	93.5%

* Computed from the reviewed article data.

GNNs excel in modeling student-course relationships (Chen et al., 2023; Li et al., 2022), while LSTMs (84.6% – 89.2%) capture temporal learning patterns (Liu et al., 2023; Waheed et al., 2023). Hybrid models consistently deliver strong performance (92.5% – 96.0%), suggesting that architectural integration effectively leverages complementary strengths.

However, substantial within-family variance across all model types underscores that architectural choice alone does not guarantee performance; rather, dataset quality, feature engineering, class balancing, and hyperparameter optimization emerge as critical determinants. Furthermore, incomplete metric reporting in several high-performing studies limits comprehensive cross-study evaluation, highlighting the need for standardized reporting practices. With tree-based models representing over half the studies in the tables and demonstrating well-established performance patterns, and linear models serving as standard baselines, the following subsections focus on architectures with specialized capabilities, novel mechanisms, or significant performance variability that merit detailed examination.

Bayesian Models

Naïve Bayes appears in only one study (Alangari and Alturki, 2020), where it achieved 91% accuracy in predicting students' final GPA. The study, which compared 15 classification algorithms, found that Naïve Bayes and Hoeffding Tree performed best, significantly outperforming other models. The dataset contained 5566 students' academic records over seven years, with feature engineering and preprocessing playing a key role in improving model performance. Despite this strong performance, the model is rarely used in student performance prediction because it assumes that features are independent. Educational data often involve correlated factors such as demographics, engagement, and prior performance, which makes other models more suitable for capturing complex relationships.

Instance-Based Models

The k-NN algorithm is a simple yet effective method, achieving an accuracy of 93%, with a high precision of 98% and a recall of 95%, as reported by Ravi et al. (2023). The study evaluated k-NN alongside other ML models in predicting student performance based on data collected from 325 B.Tech Information Technology students from Thiagarajar College of Engineering, Madurai. The model's performance is attributed to factor analysis (FA), which refined the dataset by selecting the most relevant features from 66 attributes. The results show that k-NN benefits from well-preprocessed data. However, its reliance on distance-based computations makes it less suitable for larger datasets.

Support Vector Machines (SVMs)

The performance of SVMs varies significantly, with one study reporting an accuracy of 70% (Damuluri et al., 2020), while another does not provide accuracy but reports a high precision of 96.4% and recall of 98.7% (Walid et al., 2022). This suggests that SVM can be effective, but its performance depends on factors such as kernel selection and hyperparameter tuning. The difference in results can also be due to the nature of the datasets. The lower accuracy reported by Damuluri et al. (2020) is likely due to the use of raw LMS navigational data. This may introduce noise and weak feature relationships. On the other hand, Walid et al. (2022) uses a structured dataset with academic and demographic attributes that better capture student performance patterns. Additionally, preprocessing techniques play an important role in the model performance. Walid et al. (2022) applies resampling strategies like Edited Nearest Neighbor (ENN) and borderline SVM-based SMOTE, which help balance the dataset. This shows that SVM's effectiveness is dependent on dataset characteristics, feature engineering, and class distribution. Thus, it is essential to tailor preprocessing and optimization techniques to improve the performance of the model.

Novel ML Models

A unique approach in the reviewed studies is the Random Wheel Algorithm (Khan et al., 2021a). This model achieves an accuracy of 80.5% with a high precision of 95.7%, indicating strong classification capability. This approach employs a novel mechanism based on rotating wheels to predict class labels. Each class is represented by a wheel, with forces applied based on feature combinations such as student quality and domain knowledge. These forces translate into angular velocities, where the wheel with the highest cumulative velocity across multiple trials is selected as the predicted class. The algorithm introduces a confidence measure by evaluating the gap between the winner and runner-up wheels' velocities. This approach is particularly suited for early prediction tasks, like student performance forecasting, offering both prediction and confidence metrics to inform timely interventions.

Multiple Layer Perceptrons (MLPs)

MLPs are widely used in student performance prediction due to their ability to capture complex patterns. However, their performance varies significantly depending on dataset quality and hyperparameter tuning. Reported accuracies range from 72.5% (Sun et al., 2023) to 96.4% (Aslam et al., 2021), with F1-scores reaching 96.2%. The highest performance of 96.4% accuracy) reported by (Aslam et al., 2021) was achieved by tuning hyperparameters, using feature selection techniques,

and optimizing the DL architecture. The study also employed data preprocessing strategies to improve model generalization. Despite the flexibility of MLPs, they require large datasets and fine-tuning to maintain high accuracy. Their lack of interpretability remains a challenge in educational applications.

Convolutional Neural Networks (CNNs)

Although primarily designed for image processing (Alnasyan et al., 2024a), CNNs have been applied to student performance prediction when structured patterns exist in the data. Their ability to capture spatial relationships between features makes them effective for classification tasks. In the reviewed study (Alnasyan et al., 2024a), CNNs achieved 96.6% accuracy when optimized using Binary Particle Swarm Optimization (BPSO), compared to 93.85% accuracy without feature selection. These results suggest that feature selection techniques play a crucial role in improving CNN performance by removing irrelevant or redundant data. Despite their strong predictive capabilities, CNNs require high computational resources and careful tuning of hyperparameters. Their black-box nature also limits interpretability, making them less practical for direct educational interventions where transparency is essential.

Graph Neural Networks (GNNs)

Instead of relying on traditional tabular data, GNN models transform datasets into graph structures, where students, courses, and interactions serve as nodes and edges. GNNs take into consideration dependencies that other models might overlook. Two studies report strong performance: relationship matrix-based hybrid neural network (RMHNN) achieved 93.1% accuracy and a 90.4% F1-score (Chen et al., 2023), while multi-view hypergraph neural network (MVHGNN) used hypergraph structures to model higher-order relationships, reaching an F1-score of 92.2% (Li et al., 2022).

Model performance was influenced by graph construction techniques, feature embeddings, and aggregation mechanisms. The Louvain clustering algorithm optimized student groupings (Chen et al., 2023), while cross-attention layers and hypergraph structures improved interaction modeling (Li et al., 2022). These findings suggest that GNNs are effective for student performance prediction, especially when student-course relationships are key.

Long Short-Term Memory Networks (LSTM)

LSTMs are designed for sequential data as they track student progress over time. They have demonstrated a strong performance in handling temporal dependencies. For instance, Waheed et al. (2023) achieved 84.6% accuracy by applying an LSTM model to a self-paced online learning dataset. The study found that even with data from the first five weeks, the model maintained predictive power. The early prediction ability of the model allows an early detection of students at risk. Similarly, Liu et al. (2023) explored LSTMs using clickstream data from the Open University Learning Analytics Dataset (OULAD). The model achieved an accuracy of 89.2% and an F1-score of 92.7%. These findings suggest that LSTMs are well-suited for student performance prediction, particularly for detecting at-risk students at an early stage. However, performance variations across studies indicate that dataset characteristics, feature selection, and hyperparameter tuning significantly impact model effectiveness.

Radial Basis Function Networks (RBFNs)

The study utilizing Radial Basis Function Neural Networks (RBFNs) for student performance prediction achieved 86.6% accuracy and an F1-score of 89.7% (Olabanjo et al., 2022). This model was trained on secondary school data incorporating academic, cognitive, and psychomotor scores. The results demonstrate that non-academic factors contribute to performance predictions. Principal Component Analysis (PCA) was used to optimize feature selection which contributed to improving the model performance. Despite the good performance of the model, RBFNs remain underexplored in educational contexts. Further investigation into their adaptability across different datasets could provide insights into their broader applicability.

Recurrent Neural Networks (RNN)

Turabieh et al. (2021) employ a Layered Recurrent Neural Network (LRNN) for student performance prediction, achieving an accuracy of 92%. However, the study does not provide precision, recall, or F1-score values, which makes it difficult to fully assess the model's effectiveness. The advantage of LRNN is its ability to grasp temporal dependencies in student data. This allows it to consider past performance trends when making predictions. Further exploration of RNNs in student performance prediction could provide more information about their potential for long-term forecasting.

Hybrid Models

Liu et al. (2022) propose a hybrid DL model (GDPN) that integrates Gated Recurrent Unit (GRU) and Deep Neural Network (DNN) to predict student performance. This model captures temporal learning behaviors using GRU as well as the DNN to extract cumulative behavioral features. An attention mechanism assigns weights to these feature sets to prioritize the most relevant ones. The GDPN model achieves 92.5% accuracy, which demonstrates its effectiveness in combining sequential and overall learning patterns.

Additionally, Hussain et al. (2020) present a different hybrid approach that combines multiple DL techniques and achieves a 96.0% accuracy. This model integrates hierarchical structures and optimizes feature representations to enhance predictive performance. The high accuracy suggests that hybrid models make good use of the strengths of the different architectures. Thus, hybrid models are a promising direction for improving student performance prediction.

5.4.2. Regression: Predicting continuous outcomes

Regression models aim to predict continuous numerical outcomes, such as student grades or cumulative GPA. The reviewed studies employ both ML and DL regression models, with varying performance depending on dataset characteristics and evaluation metrics.

Tree-based models are the most commonly used ML approaches for regression. Hussain and Khan (2023) combined the Genetic Algorithm (GA) and DT to achieve a Root Mean Squared Error (RMSE) of 5.34 and a Mean Absolute Error (MAE) of 11.41. The study also reported a high correlation coefficient (0.96), indicating strong predictive performance. However, no R^2 value was provided, making it difficult to assess the model's overall fit. Nachouki et al. (2023) used a Random Forest Regressor for grade prediction and reported an MAE of 7.11 on a 0–100 scale, which the authors characterized as "90.33% accuracy" by converting the error to a percentage. However, this usage of "accuracy" is unconventional for regression tasks and represents a percentage-based interpretation of MAE rather than classification accuracy. Sandoval et al. (2018) evaluated a different Random Forest model on a dataset where grades were scaled between 0 and 1, achieving an R^2 of 67.36%, an MAE of 0.0615, and an RMSE of 0.0848. These results indicate that tree-based models can effectively model student performance, but performance varies depending on data preprocessing and feature selection.

DL approaches, particularly Long Short-Term Memory (LSTM) and Graph Neural Networks (GNNs), have also been explored for regression tasks. Dien et al. (2020) applied LSTM on 16 different datasets, selecting the best MAE result from a foreign language dataset. The model achieved an MAE of 0.63115 and an RMSE of 0.79522, though no R^2 value was reported, making comparisons difficult. This study highlights the potential of LSTM in capturing temporal patterns in student performance but underscores the challenge of evaluating its effectiveness across different datasets. Lu et al. (2021) introduced a Dynamic Graph-based Time-Edge Aware Model (DGTEAM) for predicting grades using GNNs. This model incorporated three types of dynamic graphs (student-student, course-course, and student-course interactions) to improve predictions. It achieved an MAE of 7.23 and an MSE of 82.85, with an RMSE of 9.10. While these values indicate reasonable performance, they cannot be directly compared with other studies due to differences in grade scales.

A persistent methodological limitation in regression-based student performance prediction lies in the inconsistency of grading scales and reporting practices across studies. While some datasets report scores on a 0–4 scale and others on 0–100, this variation makes absolute metrics such as RMSE and MAE difficult to interpret and compare across contexts. Because these metrics are sensitive to the magnitude of the target variable, comparing results across datasets without normalization can lead to misleading impressions of relative performance. In such cases, differences in error values may reflect grading scales rather than true predictive quality. Moreover, the lack of standardized evaluation protocols limits comparability and reproducibility, making it difficult to assess progress in regression-based educational modeling.

To address these issues, future research should report normalized or relative metrics such as Mean Absolute Percentage Error (MAPE), normalized RMSE (nRMSE), or scaled MAE, which provide scale-independent assessments of proportional rather than absolute deviations. Transparent reporting should include the original grading scale, normalization steps applied, and metric definitions to enable fair cross-study comparison and reproducible benchmarking.

Overall, while regression models provide valuable insights into student performance prediction, their effectiveness is highly dependent on dataset characteristics, feature engineering, and model choice. More consistent evaluation frameworks are needed to assess their performance.

5.5. Explainability in educational data mining

5.5.1. The importance of model transparency in education

In EDM, it's not enough for predictions to be accurate, they must also be understandable. Certain models, like DTs, k-NN, and LR, provide clear insight into how they generate predictions. On the other hand, more complex models, such as ensembles and DL, often lack this transparency. These models are inherently complex and require sophisticated tools to understand their decisions.

For instance, consider using an ML model to predict whether a student might drop out of university based on their study habits and previous test scores. The model is trained on data where the outcomes (e.g., dropout or graduation) are already known, and then used to forecast results for new students. While the model might predict student retention accurately, it often fails to clarify the factors influencing each prediction. This lack of transparency can make decision-makers hesitant to rely solely on the model's outputs, as the underlying logic remains obscure.

There is a trade-off between a model's predictive power and its interpretability. Those that are highly performant may be harder to interpret, and vice versa as shown in Fig. 11. In education, where such predictions can heavily influence decisions, understanding the model's reasoning is vital. This is why XAI tools are increasingly employed to make these complex models more transparent and interpretable.

White-box models, such as DTs, k-NN, and Logistic Regression (Abuzinadah et al., 2023), offer a straightforward understanding of how inputs are transformed into outputs. This transparency makes them particularly valuable in educational contexts, where stakeholders including instructors, students, and administrators need to understand and trust model outcomes before acting upon them. However, as learning environments become increasingly data-rich, the use of more sophisticated models has grown, creating a need to balance accuracy with interpretability.

5.5.2. Explaining black-box models: insights for educational practice

Explainability was significantly less frequent with black-box models. Only 6 studies (14% of those employing neural or ensemble models) applied post-hoc XAI techniques: SHAP in 5 studies (Jang et al., 2022; Karlos et al., 2020; Sahlaoui et al., 2021; Waheed et al., 2023; Wang et al., 2023) and LIME in one (Dien et al., 2020). Although other post-hoc techniques like Anchors and Integrated Gradients are widely used in broader XAI research, none were applied in the reviewed studies, indicating SHAP and LIME currently dominate EDM practice.

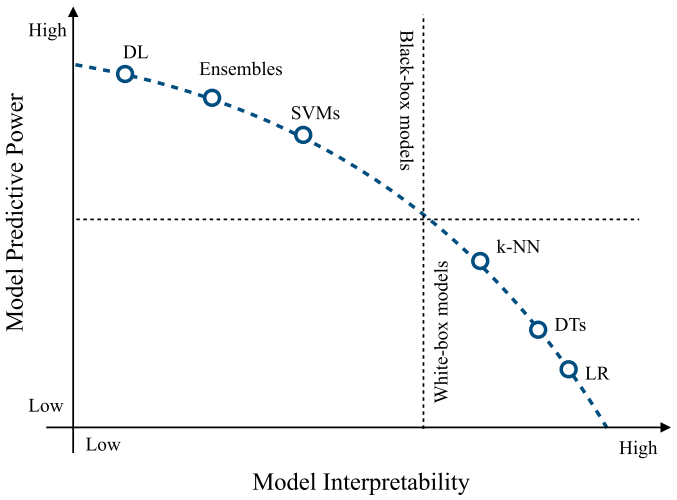


Fig. 11. Predictive power vs. interpretability trade-off.

Despite limited adoption, consistent insights emerged. SHAP analyses revealed behavioral engagement (e.g., clicks, assessment access, platform navigation) and continuous evaluation performance as the strongest determinants of student success or risk (Karlos et al., 2020; Sahlaoui et al., 2021; Wang et al., 2023). However, multiple works showed high engagement doesn't universally correspond to positive outcomes (Jang et al., 2022; Waheed et al., 2023), as students with similar activity levels may exhibit different performance trajectories. Such variability indicates that global importance rankings may obscure individual challenges, reinforcing the need for localized explanations.

LIME provided localized interpretability for DL regression models by showing why predictions were accurate or erroneous (Dien et al., 2020). Even strong predictors like CGPA positively contributed to accurate predictions in some cases while acting as negative contributors in mispredicted cases, revealing predictive instability at the individual level. This highlights that population-level signals may not reliably generalize to every student and underscores the need for explainability to validate risk assessments before applying interventions. From a pedagogical standpoint, failing to identify a struggling learner is more serious than offering unnecessary support. Local explanations provide a safeguard for proactive yet responsible intervention.

Fig. 12 illustrates local SHAP explanations, showing why a specific student was classified as at-risk. Red features increase failure likelihood (e.g., low homework grades, limited peer connections), while blue features reduce it (e.g., strong performance on specific assessments)(Jang et al., 2022). Such explanations help educators tailor targeted support strategies. This demonstrates that interventions shouldn't rely solely on global feature rankings: students with the same predicted risk may struggle for different reasons. Local explanations shift focus from one-size-fits-all remediation toward learner-centered support addressing individual barriers.

Wang et al. (2023) demonstrated that explainability complements model robustness: SHAP-based analysis identified both academic history and behavioral data as performance prediction drivers, while emphasizing that interpretability alone doesn't mitigate challenges such as data imbalance. Collectively, these studies suggest post-hoc XAI exposes both strengths and limitations of predictive models, enabling justified and cautious interpretation of outputs.

5.5.3. The educational value of XAI: From transparency to trust

Building upon the explainability typology discussed previously, this section examines how transparency translates into educational value. Explainability in EDM bridges computational insight and pedagogical

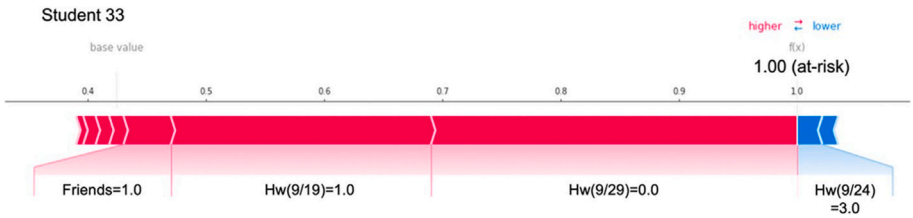


Fig. 12. Local SHAP explanation showing feature contributions for an at-risk student (Jang et al., 2022).

understanding (Khosravi et al., 2022), enabling educators to align analytics with instructional design and interventions.

The educational value of XAI: From transparency to trust. XAI methods reveal factors influencing student success or risk predictions (Khosravi et al., 2022). Interpretable models directly display relationships (e.g., how attendance affects outcomes), while post-hoc methods like SHAP or LIME explain feature contributions in complex models (Mustofa et al., 2025; Tiukhova et al., 2024). This enables data-informed reflection and pedagogical adaptation. Transparent models foster trust, allowing teachers to interpret outputs within the classroom context and transform statistical information into actionable guidance (Khosravi et al., 2022).

However, the transparency-trust relationship is nuanced: contextualized explanations are more actionable than technical feature rankings. Effective translation requires collaboration between data scientists and pedagogical experts. Institutions are increasingly establishing analytics teams or partnerships to bridge computational complexity and educational practice (Khushalani, 2025). When explanation design, professional development, and institutional support are in place, explainability mediates between analytics and human expertise, fostering evidence-informed yet context-sensitive teaching.

Pedagogical innovation through predictive and explainable AI. Combined predictive and XAI represent significant pedagogical innovation. Predictive analytics enable real-time support for at-risk students, while explainability ensures predictions are transparent and pedagogically grounded. Together, they transform educational data into a dynamic, feedback-driven system. This empowers educators to align insights with instructional goals, personalize interventions, and foster learner self-awareness, redefining how evidence guides teaching and learning.

Implementation considerations: Ensuring validity and equity. Integration enables proactive, data-informed monitoring (Khosravi et al., 2022; Tiukhova et al., 2024), but requires careful attention to validity and equity. Stability and generalizability of explanations are critical. XAI techniques identify features in specific datasets, but relationships may reflect context-dependent patterns rather than universal principles. Most studies applied XAI within single institutions, highlighting the need for cross-institutional validation. Feature-based explanations predominantly highlight measurable variables like clicks, time-on-task, or assessment performance (Karlos et al., 2020; Liu et al., 2022; Xu and Qu, 2021), while affective, relational, and socio-cultural factors remain underexplored. Complementing quantitative insights with qualitative perspectives yields holistic understanding and promotes equitable interventions.

Connections to educational theory. Factors identified by explainable models, including engagement (Liu et al., 2022; Xu and Qu, 2021), participation, and assessment behavior (Karlos et al., 2020), align with established frameworks such as formative assessment principles, self-regulated learning theory (Brenner, 2022), and constructivist approaches (Do et al., 2023). XAI reinforces established principles while uncovering subtle relationships among learning behaviors. By capturing diverse learning trajectories, XAI supports inclusive environments, revealing variations across learners with different backgrounds and needs. By enabling real-time monitoring, XAI operationalizes educational theory, transforming constructs into actionable interventions.

Directions for advancing the field. To maximize XAI's educational value, key directions include: (1) empirical studies on how educators interpret and use explanations; (2) investigation of explanation designs bridging computational reasoning and pedagogical understanding; (3) evaluation of whether explainability improvements lead to learning gains; and (4) attention to equity, bias mitigation, and transparency. Advancing these directions ensures AI integration represents genuine pedagogical advancement, strengthening the partnership between human judgment and intelligent systems.

6. Conclusion

This systematic review examined 72 peer-reviewed studies (2017-2024), synthesizing ML, DL, and XAI applications in student performance prediction. By analyzing these paradigms together, this work reveals that advancing predictive accuracy requires integrating technical innovation with pedagogical alignment and institutional feasibility.

The analysis shows no single algorithm consistently outperforms others across all datasets. While DL models achieve high classification accuracy, traditional ML models remain strong competitors due to interpretability and lower computational requirements. Model selection must be guided by institutional context, data characteristics, and pedagogical objectives.

A key finding is the widespread deployment of classification over regression models. This reflects institutions' practical need to identify at-risk students for timely intervention. Binary classification tasks, such as dropout prediction, are most prevalent, emphasizing early intervention strategies. This aligns with proactive, student-centered institutional analytics goals.

Another crucial factor is the accuracy-interpretability trade-off. While DL models offer high performance, their lack of transparency poses deployment challenges. Only 14% of studies using black-box models applied post-hoc XAI techniques. Explainability techniques like SHAP and LIME provide insights into model decisions. However, limited XAI adoption represents a gap between technical capability and practical implementation, which is essential for building educator trust and ensuring actionable pedagogical interventions.

Scalability is important, as models need to maintain performance with increasing data. However, most studies focus only on predictor importance and accuracy, not scalability. Furthermore, models are usually tested on single datasets, limiting robustness evaluation. This single-institution focus limits generalizability and underscores the need for cross-institutional validation. Exploring performance across multiple datasets could provide valuable insights. Investigating how engagement patterns and behavioral features vary across contexts would enhance understanding of universal learning principles versus dataset-specific patterns.

Moving forward, hybrid approaches combining ML and DL models present a promising direction, balancing performance and interpretability while reducing computational load. Additionally, AutoML and transfer learning integration may enhance predictive capabilities across diverse educational contexts. Future research should prioritize interdisciplinary collaboration between data scientists and educators to align model development with pedagogical theory. Establishing

frameworks for responsible AI deployment, including fairness, bias mitigation, and transparent communication of limitations, will be critical.

Ultimately, this review highlights AI-driven student performance prediction's potential in personalized learning and early intervention. This work provides both a methodological reference for researchers and a strategic foundation for practitioners and policymakers. Addressing challenges related to fairness, transparency, scalability, and real-world applicability may contribute to broader adoption in educational settings.

Ethics and data availability statement

This review uses only published peer-reviewed studies. No new data were collected. All data are available in the cited sources and from the corresponding author upon request.

CRediT authorship contribution statement

Salma Boujmiraz: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Hassane Darhmaoui:** Writing – review & editing, Supervision, Conceptualization. **Ahmed Drissi el maliani:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table 7
Detailed references for machine learning models used in the reviewed studies.

Model sub-type	References
Multiple linear regression	(Dabhade et al., 2021; Karlos et al., 2020; Kostopoulos et al., 2019; Sandoval et al., 2018)
Logistic regression	(Abuzinadah et al., 2023; Alachiotis et al., 2022; Alcaraz et al., 2021; Alhazmi and Sheneamer, 2023; Hoffait and Schyns, 2017; Injadat et al., 2020a,b; Jang et al., 2022; Jasim et al., 2021; Karlos et al., 2020; Li et al., 2022; Liu et al., 2023; Mastour et al., 2023; Pek et al., 2023; Riestra-González et al., 2021; Sandoval et al., 2018; Sultana et al., 2017; Sun et al., 2023; Tamada et al., 2022; Trakunphutthirak and Lee, 2022; Waheed et al., 2020, 2023; Walid et al., 2022; Xu and Qu, 2021; Yousafzai et al., 2021; Zeineddine et al., 2021)
Decision trees	(Alachiotis et al., 2022; Alangari and Alturki, 2020; Alcaraz et al., 2021; Cheng et al., 2024; Costa et al., 2017; Dixit et al., 2022; Esteban et al., 2021; Helal et al., 2018; Hussain and Khan, 2023; Imran et al., 2019; Jang et al., 2022; Jasim et al., 2021; Jenkins et al., 2022; Khan et al., 2021b; Kostopoulos et al., 2019; Li et al., 2022; Pek et al., 2023; Riestra-González et al., 2021; Rizvi et al., 2019; Rodríguez-Hernández et al., 2021; Sahlaoui et al., 2021; Sultana et al., 2017; Sun et al., 2023; Tamada et al., 2022; Tsiakmaki et al., 2020, 2021; Waheed et al., 2023; Xu and Qu, 2021; Yousafzai et al., 2020; Zeineddine et al., 2021)
Random forests	(Abuzinadah et al., 2023; Al-Zawqari et al., 2022; Alangari and Alturki, 2020; Alcaraz et al., 2021; Alhazmi and Sheneamer, 2023; Cheng et al., 2024; Dixit et al., 2022; Esteban et al., 2021; Fan et al., 2023; Hoffait and Schyns, 2017; Hung et al., 2020; Injadat et al., 2020a,b; Jang et al., 2022; Jasim et al., 2021; Jenkins et al., 2022; Khan et al., 2021a,b; Kostopoulos et al., 2019; Lemay and Doleck, 2020; Li et al., 2022; Liu et al., 2023; Mastour et al., 2023; Nachouki et al., 2023; Pek et al., 2023; Rodríguez-Hernández et al., 2021; Sandoval et al., 2018; Tamada et al., 2022; Trakunphutthirak and Lee, 2022; Tsiakmaki et al., 2020; Walid et al., 2022; Wang et al., 2023; Xue and Niu, 2023; Yousafzai et al., 2021)
Gradient boosting machines	(Abuzinadah et al., 2023; Alcaraz et al., 2021; Alhazmi and Sheneamer, 2023; Begum and Padmannavar, 2022; Cheng et al., 2024; Esteban et al., 2021; Fan et al., 2023; Jang et al., 2022; Jasim et al., 2021; Jenkins et al., 2022; Karlos et al., 2020; Khan et al., 2021a; Li et al., 2022; Liu et al., 2023; Mastour et al., 2023; Niyogisubizo et al., 2022; Pek et al., 2023; Rodríguez-Hernández et al., 2021; Sahlaoui et al., 2021; Tamada et al., 2022; Trakunphutthirak and Lee, 2022; Waheed et al., 2023; Walid et al., 2022; Wang et al., 2023; Xue and Niu, 2023)
Support vector machine	(Alachiotis et al., 2022; Alcaraz et al., 2021; Alhazmi and Sheneamer, 2023; Costa et al., 2017; Damuluri et al., 2020; Dixit et al., 2022; Fan et al., 2023; Hung et al., 2020; Injadat et al., 2020a,b; Jang et al., 2022; Jasim et al., 2021; Khan et al., 2021a,b; Li et al., 2022; Mastour et al., 2023; Ravi et al., 2023; Riestra-González et al., 2021; Tamada et al., 2022; Waheed et al., 2020, 2023; Walid et al., 2022; Wang et al., 2023; Yousafzai et al., 2021; Zeineddine et al., 2021)
Support vector regression	(Dabhade et al., 2021)
Naive bayes classifier	(Abuzinadah et al., 2023; Alachiotis et al., 2022; Alangari and Alturki, 2020; Alcaraz et al., 2021; Costa et al., 2017; Esteban et al., 2021; Helal et al., 2018; Injadat et al., 2020a,b; Jasim et al., 2021; Khan et al., 2021a; Lemay and Doleck, 2020; Nabizadeh et al., 2022; Pek et al., 2023; Ravi et al., 2023; Riestra-González et al., 2021; Sultana et al., 2017; Tamada et al., 2022; Tsiakmaki et al., 2020; Turabieh et al., 2021; Zeineddine et al., 2021)
k-nearest neighbour	(Alachiotis et al., 2022; Alcaraz et al., 2021; Alhazmi and Sheneamer, 2023; Cheng et al., 2024; Dixit et al., 2022; Hussain and Khan, 2023; Injadat et al., 2020a,b; Jasim et al., 2021; Karlos et al., 2020; Khan et al., 2021a; Li et al., 2022; Liu et al., 2023; Mastour et al., 2023; Nabizadeh et al., 2022; Pek et al., 2023; Priyambada et al., 2023; Ravi et al., 2023; Trakunphutthirak and Lee, 2022; Turabieh et al., 2021; Waheed et al., 2023; Walid et al., 2022; Xu and Qu, 2021; Yousafzai et al., 2020, 2021; Zeineddine et al., 2021)

Table 8
Classification of features used by each study (temporal vs. non-temporal).

Reference	Temporal features		Non-temporal features		
	Acad.	Behavioral	Demogr.	Acad. Background	Other
(Abuzinadah et al., 2023)		✓	✓	✓	
(Alachiotis et al., 2022)	✓	✓	✓	✓	
(Alangari and Alturki, 2020)	✓		✓	✓	
(Alcaraz et al., 2021)	✓		✓	✓	
(Alhazmi and Sheneamer, 2023)	✓	✓	✓		
(Al-Zawqari et al., 2022)	✓	✓			
(Aslam et al., 2021)		✓	✓	✓	
(Begum and Padmannavar, 2022)	✓	✓	✓	✓	
(Alnasyan et al., 2024a)	✓		✓	✓	✓
(Chen et al., 2023)	✓		✓	✓	✓
(Cheng et al., 2024)	✓		✓	✓	✓
(Costa et al., 2017)	✓	✓	✓	✓	✓
(Dabhade et al., 2021)	✓	✓	✓	✓	✓
(Damuluri et al., 2020)	✓	✓			
(Dien et al., 2020)	✓		✓	✓	✓
(Dixit et al., 2022)	✓		✓	✓	✓
(Esteban et al., 2021)	✓			✓	
(Fan et al., 2023)	✓	✓	✓	✓	✓
(Giannakas et al., 2021)	✓	✓	✓		✓
(Helal et al., 2018)		✓	✓	✓	
(Hoffait and Schyns, 2017)			✓	✓	✓
(Hung et al., 2020)		✓	✓		✓
(Hussain et al., 2020)		✓	✓	✓	✓
(Hussain and Khan, 2023)	✓			✓	
(Imran et al., 2019)	✓	✓	✓	✓	✓
(Injadat et al., 2020a)	✓				
(Injadat et al., 2020b)	✓				
(Jang et al., 2022)	✓	✓			✓
(Jasim et al., 2021)		✓	✓	✓	✓
(Jenkins et al., 2022)	✓	✓	✓	✓	✓
(Karlos et al., 2020)	✓	✓	✓	✓	
(Khan et al., 2021a)				✓	✓
(Khan et al., 2021b)	✓	✓	✓	✓	✓
(Lemay and Doleck, 2020)	✓	✓			
(Li et al., 2022)	✓	✓	✓	✓	✓
(Liu et al., 2022)	✓	✓			
(Liu et al., 2023)	✓	✓			
(Lu et al., 2021)	✓			✓	✓
(Mastour et al., 2023)	✓		✓	✓	✓
(Nabizadeh et al., 2022)	✓	✓	✓		✓
(Nachouki et al., 2023)		✓	✓	✓	✓
(Niyogisubizo et al., 2022)	✓	✓			
(Olabanjo et al., 2022)	✓	✓		✓	
(Pek et al., 2023)	✓		✓	✓	✓
(Priyambada et al., 2023)	✓	✓			✓
(Ravi et al., 2023)		✓	✓		✓
(Ranjeeth et al., 2021)	✓	✓	✓	✓	✓
(Riestra-González et al., 2021)	✓	✓			✓
(Rizvi et al., 2019)			✓	✓	
(Rodríguez-Hernández et al., 2021)	✓		✓	✓	✓
(Sahlaoui et al., 2021)	✓	✓	✓	✓	✓
(Sandoval et al., 2018)	✓	✓	✓	✓	
(Sultana et al., 2017)	✓	✓	✓	✓	✓
(Tamada et al., 2022)	✓	✓	✓		
(Trakunphutthirak and Lee, 2022)	✓	✓	✓	✓	✓
(Tsiakmaki et al., 2020)	✓	✓	✓		
(Tsiakmaki et al., 2021)	✓	✓	✓		
(Turabieh et al., 2021)	✓	✓	✓	✓	✓
(Waheed et al., 2020)	✓	✓	✓	✓	
(Waheed et al., 2023)	✓	✓	✓		
(Walid et al., 2022)	✓	✓	✓		✓
(Wang et al., 2023)	✓	✓		✓	✓
(Xu and Qu, 2021)		✓			✓
(Xue and Niu, 2023)	✓	✓		✓	
(Yang and Bai, 2022)	✓	✓		✓	
(Yousafzai et al., 2021)	✓		✓	✓	✓
(Yousafzai et al., 2020)	✓				
(Yurtkan et al., 2023)		✓	✓	✓	
(Zeineddine et al., 2021)			✓	✓	✓

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